Final Executive Summary Document Content

Project Code: XY-ML-P1

Project Title: Strategic Market Valuation Modeling via Advanced Regression Architectures

Lead AI Engineer: Prashant Rai

1. Architectural Diagnostics & Metadata

- **Dataset Volumetrics:** The structural dimension consists of 545 discrete residential rows mapped across 13 native categorical and numerical variables.
- **Feature Matrices:** Non-numeric markers (mainroad, airconditioning, furnishingstatus, etc.) were normalized using one-hot encoding schemas, expanding the training framework into a highly deterministic 14-dimensional feature matrix.
- **System Preprocessing:** Features were mapped to an identical scale using standard Z-score normalization (StandardScaler) to smooth computational descent curves.

2. Analytical Comparison of Models

The modeling pipeline evaluated a parametric Ordinary Least Squares (OLS) Linear Regression baseline against an ensemble Random Forest architecture with max-depth constraints to check generalization bounds:

Evaluation Metric	Baseline Linear Regression	Tuned Random Forest Regressor
Mean Absolute Error (MAE)	**₹9,70,043.40**	₹10,19,373.72
Root Mean Squared Error (RMSE)	**₹13,24,506.96**	₹14,03,691.56
R^2 Score (Variance Capture)	**0.6529 (65.29%)**	0.6102 (61.02%)

3. Core Analytical Inferences

- **Primary Pricing Anchors:** Feature weight coefficients confirm that total square footage (**area** at 0.54 correlation) combined with functional layouts like **bathrooms**

(0.52 correlation) and structural expansion (**stories** at 0.42 correlation) dictate dominant property premiums.

- **The Data Anomaly:** Interestingly, niche structural additions like full **basements** (0.19) or specialized installations like internal **hot water heating** systems (0.09) display heavily diminished marginal correlation profiles compared to macro space properties. Simple vertical configurations yield higher valuations than peripheral spatial utilities.
- **Architecture Anomaly:** Linear Regression explicitly out-performed the non-linear ensemble system by **~4.27%** in variance translation. This pattern formally establishes that price curves in this ecosystem maintain high mathematical linearity, where complex decision-boundary trees suffer overfitting bounds due to small sample configurations.

4. Enterprise Real Estate Guidance

Real estate enterprises seeking maximum margin realization should bypass high-expenditure utility retrofits (such as basement partitioning or isolated guest spaces). Instead, resource deployment channels must focus directly on capital development paths emphasizing structural plot footprint expansions (area) and vertical optimization architectures (stories and multi-bathroom setups), where the consumer valuation ceiling tracks at its peak.